University of New Haven

TAGLIATELA
COLLEGE OF ENGINEERING

UNIVERSITY CAMPUS AREA NETWORK

Enterprise Network Design

CSCI-6649-01

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Introduction:

The University Campus Area Network (UCAN) is inspired by the University of New Haven's network, using a Top-Down Design to address campus needs. It integrates Firewall, IDS/IPS, DMZ, and both wired and wireless LANs for seamless connectivity among departments and buildings. A Disaster Recovery Plan ensures resilience and continuity.

The project focuses on scalability, network management, and maintenance, ensuring the network can handle growth, is easy to manage, and performs well over time.

Technical Goals:

- **Performance:**
- High-speed data transfer rates.
- Low latency.
- Reliable network connectivity.
- **Interoperability:**
- Compatibility with various devices and systems.
- Seamless integration with existing infrastructure.
- Support for different protocols and standards.
- **Flexibility:**
- Adaptability to changing needs and technologies.
- Modular network design.
- Easy configuration and management.

Business Goals:

- **Reliability:** The aim of this network is to provide
- High uptime and minimal downtime.
- Redundancy and failover mechanisms.
- Regular maintenance and monitoring.
- **Security:**
- Protection against cyber threats (e.g., hacking, malware).
- Strong access control and authentication.
- Regular security audits and vulnerability assessments.
- **Scalability:**
- Ability to accommodate future growth and expansion.
- Flexible network infrastructure.
- Easy addition of new devices and users.

- **Cost-Effectiveness:**
- Efficient use of resources.
- Minimizing operational costs.
- Optimizing network performance.

Logical Design:

<u>Overall Network Diagram</u>: The network is designed to accommodate internet and connectivity services to 2 campuses. This is accompanied by having internet delivered to both campuses via an Internet Service Provider (ISP). This network follows the Top-Down network design architecture with focus on scalability and efficiency. Along with this, there is also implementation of Disaster Recovery Plans and security measures for safeguarding the incoming and outgoing traffic.

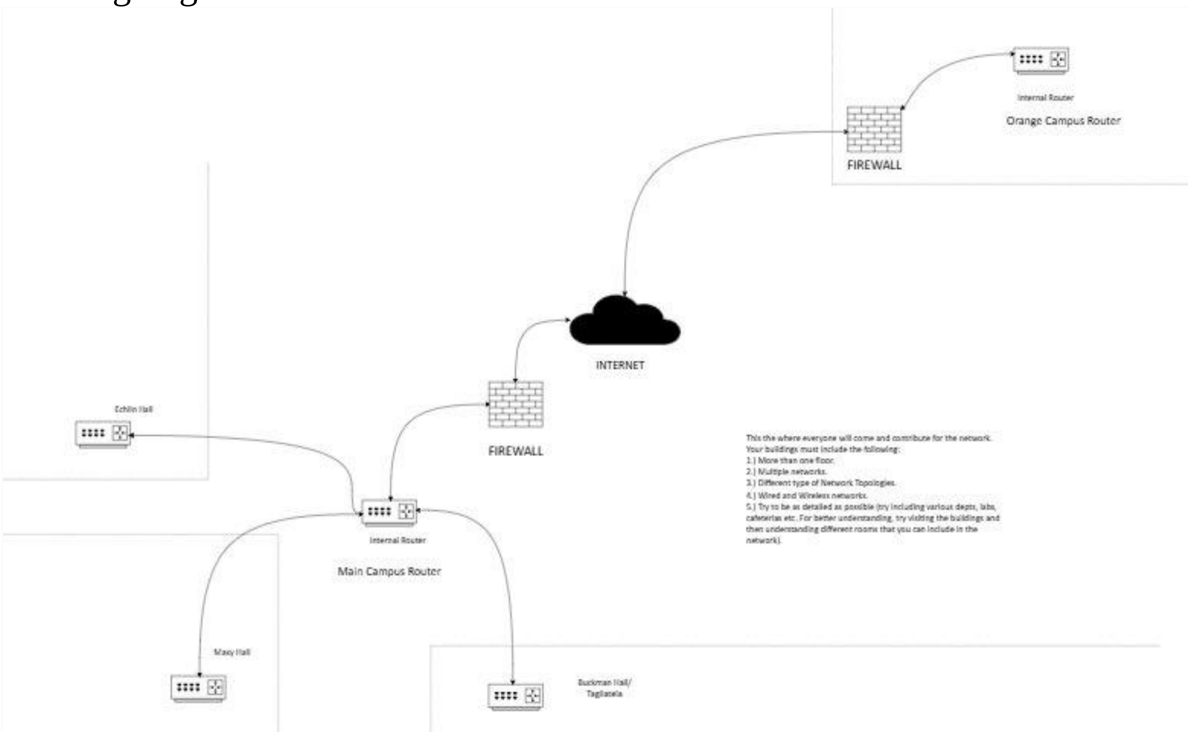


Image: Overview of the University Campus Area Network

Connection between the two Campuses:

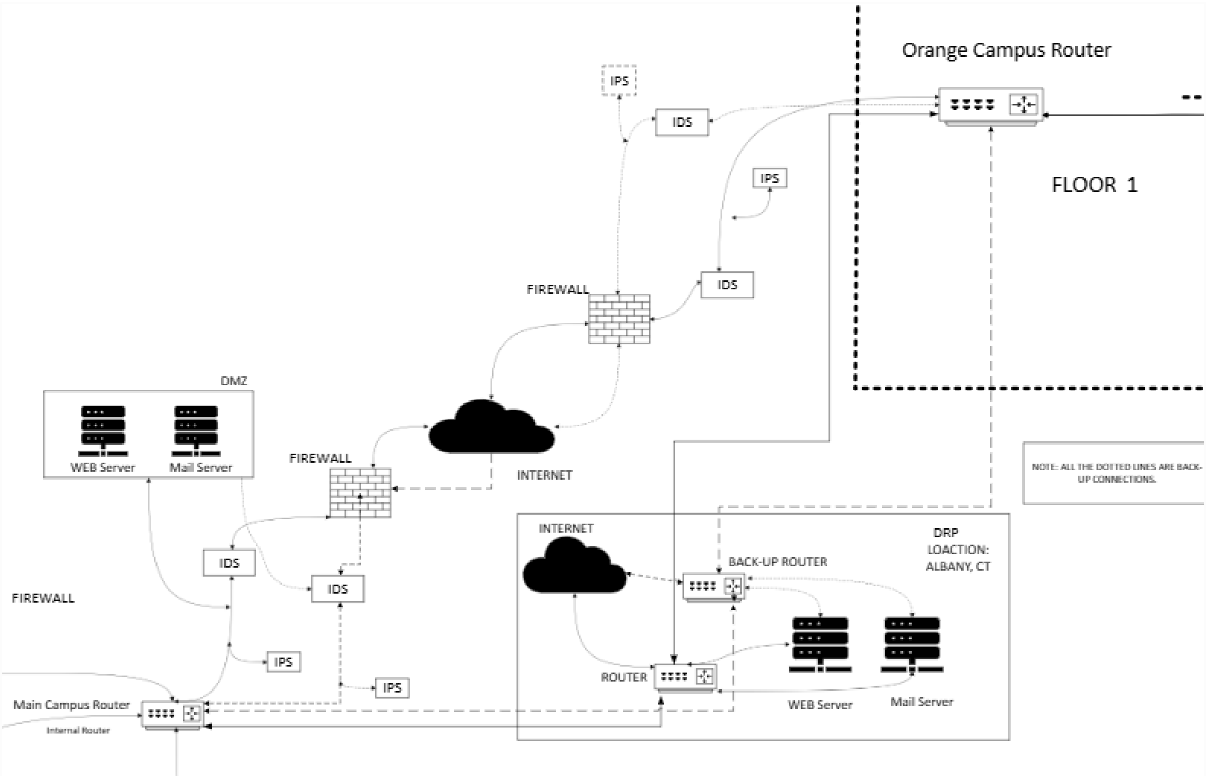


Image : Connection between the two campuses, DRP, backup, and security systems.

This section of the diagram consists of the connection between the two campuses. Here, defensive measures like firewalls, IDS and IPS are employed to protect the end user from malicious traffic. To protect the mail server and the web server of the university they are kept in De-Militarized Zone (DMZ). This is followed across the main campus and a similar approach is seen in the orange campus. The key difference in the two is that Orange Campus network does not have web server and mail server as that requirement is satisfied on the Main campus's web and mail servers.
Here, all the main routers, on both the campuses, that are responsible for routing the data are connected to a Disaster Recovery Plan (DRP). This DRP, has a Web server and a mail server. All the important data that is required for backup is backed-up at the DRP. To add to this, there is also a backup system in this architecture. The dotted lines on the diagram represent the backup systems.

The back-up systems are responsible for becoming live if the main system fails because of any problem. The backup system also has IDS and IPS connected to it for maintaining the security standards and works the same as the main system works. The backup system is also connected to the DRP and is activated in case of main system's failure.

Budget for the Connection between the two Campuses:

Column1	Column2	Column3	Column4	Column5	Column6
Components	Manufacturer	Model	Cost (appx)	No. of units	Total cost
Firewall (Main Campus)	Palo Alto Networks	PA-3220	$13500	1	$13500
Firewall (Orange Campus)	Palo Alto Networks	PA-3220	$13500	1	$13500
Router (Main Campus)	Cisco	ASR 1001-X	$10000	1	$10000
Router (Orange Campus)	Cisco	Cisco ISR 4451-X	$2500	1	$2500
Application Server (Main Campus and DRP)	Dell	PowerEdge R750	$8000	2	$16000
Mail Server (Main Campus and DRP)	HPE	ProLiant DL380 Gen11	$9000	2	$18000
IDS and IPS (Main Campus)	Cisco	Firepower 1140	$9000	1	$9000
IDS and IPS (Orange Campus)	Cisco	Firepower 1120	$9000	1	$9000
IDS and IPS (Main Campus backup)	Cisco	Firepower 1140	$9000	1	$9000
IDS and IPS (Orange Campus backup)	Cisco	Firepower 1120	$9000	1	$9000
Cables (Main Campus)	OptoSpan	PFPF-JM312NXRCD	$2.04/feet	2000 ft	$4080
Cables (Orange Campus)	OptoSpan	PFPF-JM312NXRCD	$2.04/feet	1000 ft	$2040
Cables (DRP)	CommScope	Cat 6 Ethernet Cable	0.4/feet	157 miles	$331584
Cables (Main Campus backup)	OptoSpan	PFPF-JM312NXRCD	$2.04/feet	2000 ft	$4080
Cables (Orange Campus backup)	OptoSpan	PFPF-JM312NXRCD	$2.04/feet	1000 ft	$2040
Application Server (Backup)	Dell	PowerEdge R750	$8000	1	$8000
Mail Server (Backup)	HPE	ProLiant DL380 Gen11	$9000	1	$9000
Cables (DRP backup)	CommScope	Cat 6 Ethernet Cable	0.4/feet	157 miles	$331584
Total					$801608

For the Main Campus and Orange Campus Palo Alto's PA-3220 is used for firewall. This device provides high performance to a medium sized enterprise and is capable of advanced threat protection.

For Router on the Main Campus, Cisco's ASR 1001-X is used. This is a router that is able to provide high scalability and has high throughput and is able to provide data without compromising performance to up to 2000 users.
The router choice for orange campus is Cisco's ISR 4451-X. This is router that has less router power than ASR 1001-X as Orange campus has fewer users than Main Campus. This router is capable of providing connectivity to up to 1500 users without any compromises on the performance.

For DRP and Main Campus's Web Server DELL's PowerEdge R750 is used. It provides reliable and powerful hardware, has virtualization support and is energy efficient which makes it cost efficient over a long period of time.
For DRP and Main Campus's Mail Server HPE's ProLiant DL380 Gen11 is used. This device is versatile and secure, provides strong performance for email hosting and redundance options.

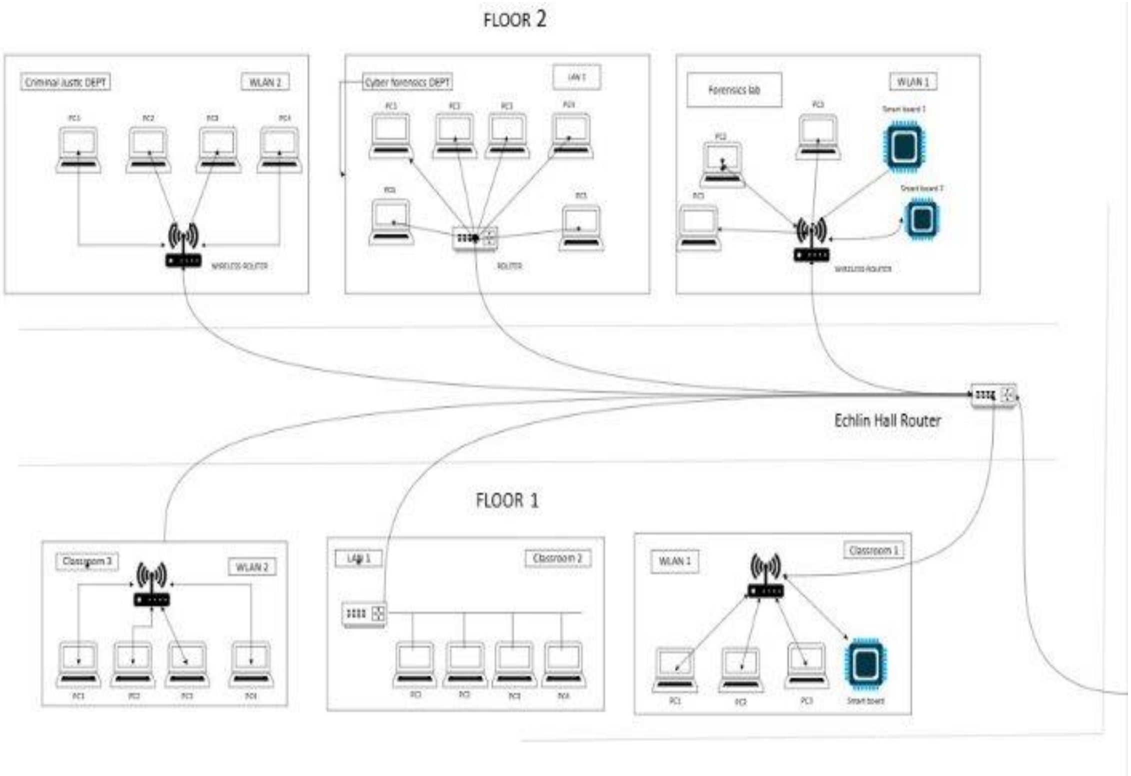
For IDS and IPS of the Main Campus and Back up, Cisco's Firepower 1140 is used. This system has both IDS and IPS in one unit and is cost effective with robust intrusion detection.

For IDS and IPS on the Orange Campus and Back up, Cisco's Firepower 1120 is used. It provides similar features to Firepower 1140 but is made for smaller traffic size networks.

For Cabling on Main and Orange Campus and their Back up systems, OptoSpan's PPF-M3121NXRCD, which is a high quality, high-capacity fiber optic cable, is used. This cable is long lasting and durable and is cost efficient when bought in bulk.

For the DRP and it's Back up system, CommScope's Cat-6 Ether net cable is used as it is affordable, reliable over long distances and costs less than a fiber optic cable over a long distance. This system may require a repeater to restore the quality of analog signals over a long distance.

ECHLIN HALL:



Building overview:

Name: Echlin Hall

The building consists of two floors, each with a mix of classrooms, labs, and departments. The building supports academic and research activities, focusing on criminal justice, cyber forensics, and general classrooms.

Network components: In this building we used two types of routers wired and wireless routers (WLAN & LAN) WLAN is used to provide wireless local area network access. And LAN routers are used to connect PCs and other wired devices to a local area network. Wireless network WLAN1 serves devices in classroom 1 and the forensics lab & WLAN2 serves devices in classroom 3 and the criminal justice department. Pcs are used for general computing, academic, and research purposes. And smart boards are located in select classrooms and labs to support teaching and collaboration.

Floor details:

The building consists of two floors:

In floor 1 we have three classrooms.

Classroom 1	Classroom 2	Classroom 3
Devices: 3 PCs & 1 smart board. **Wireless network:** connected to WLAN 1Via a wireless router.	**Devices:** 4PCs. **Wired network:** Connected through LAN 1 via a wired router.	**Devices:** 4PCs. **Wireless network:** Connected through WLAN 2 via a wireless router.

In floor 2 we have two departments and one lab.

Criminal Justice Department	Cyber Forensics Department	Forensics Lab
Devices: 4PCs. **Wireless network:** Connected to WLAN 2 via a wireless router.	**Devices:** 5Pcs. **Wired network:** All Pcs are connected via an internal switch to a wired router.	**Devices:** 2PCs & 2 smart Board. **Wireless network:** Connected to WLAN 1 Via a wireless router.

Echlin Hall router acts as the main aggregation point for all floor routers and ensures connectivity between departments, classrooms, and labs, as well as external network.

The switch connected to the Echlin Hall router likely enables external connectivity, providing access to the campus wide network or the internet.

The Criminal Justice Department is likely used for online legal research, simulations, and educational activities requiring reliable wireless access.

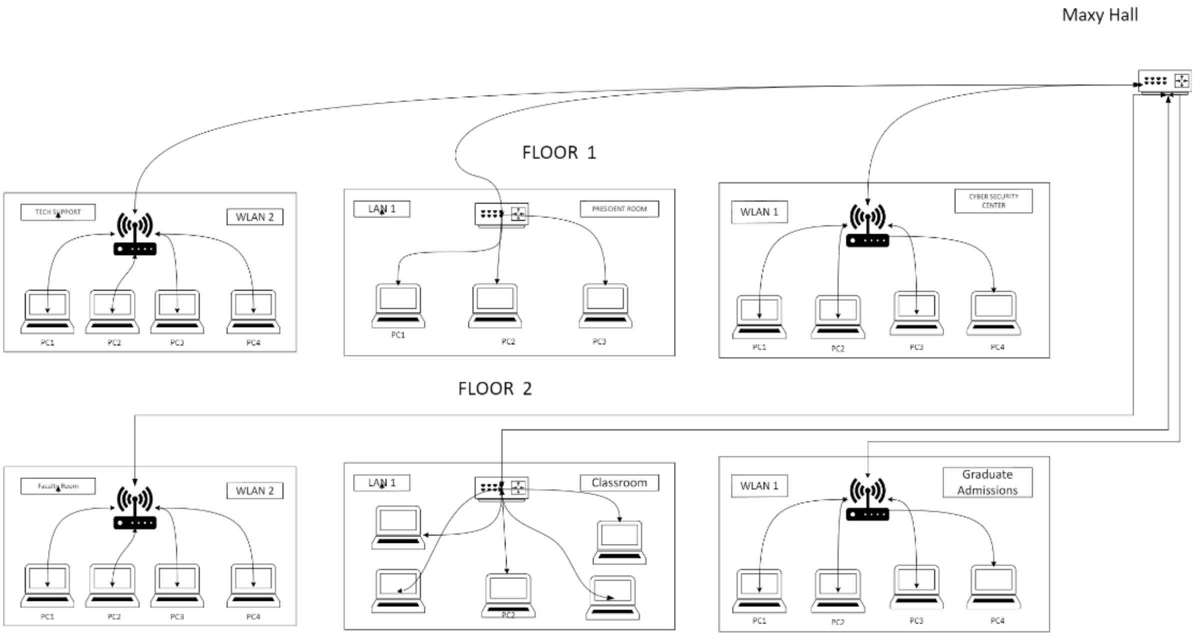
The Cyber Forensics Department focuses on data analysis, forensic investigations, high performance computing, necessitating a stable wired LAN connection.

Forensics lab equipped with smart boards for teaching or collaborative research and analysis.

Budget for the Echlin Hall:

	A	B	C	D	E	F
1	**Device**	**Brand**	**Model**	**Cost (approx)**	**No. of unit**	**Total Cost**
2	PC	Apple	iMac i7	$1245	25	$1245
3	PC	HP	Envy	$899	15	$899
4	PC	DELL	XPS	$899	20	$899
5	PC	Apple	iMac i5	$1199	15	$1199
6	PC	Lenovo	ThinkPad	$879	22	$879
7	PC	Lenovo	Legion	$879	18	$879
8	PC	Lenovo	IdeaPad	$829	13	$829
9	PC	Apple	iMac i7	$1245	23	$1245
10	PC	HP	Pavilion	$919	31	$919
11	PC	DELL	XPS	$899	11	$899
12	Smartboard	Vibe	S155	$2899	2	$2899
13	Smartboard	Vibe	S155	$2899	2	$2899
14	Router(Classroom 1)	Cisco	C9100-8P-2G-L	$1102	1	$1102
15	Router(Classroom 2)	Cisco	C9100-8P-2G-L	$1102	1	$1102
16	Router(Classroom 3)	Cisco	C9100-8P-2G-L	$1102	1	$1102
17	Router(Criminal Justice Department)	Cisco	C9100-8P-2G-L	$1102	1	$1102
18	Router(Cyber Forensics Department)	Cisco	C9100-8P-2G-L	$1102	1	$1102
19	Router(Forensics Lab)	Cisco	C9100-8P-2G-L	$1102	1	$1102
20	Router(Echlin Hall)	Cisco	C9100-8P-2G-L	$1102	1	$1102
21	Cables (Ethernet (Cat6 Cable)		Belkin Cat6 UTP Cable (bulk)	$18(0.30 per meter)	60 meters	$18
22	Cables (Fiber Optic Cable)		Corning Single-Mode Fiber Cable	$50(2.00 per meter)	25 meters	$50
23	**TOTAL**			**$23472**		**$23472**

MAXY HALL:



AN OVERVIEW OF MAXYHALL

- **Maxy Hall** serves as the main network hub. It houses a **switch** that interconnects all rooms on Floor 1 through Ethernet cables.

FLOOR 1

TECH SUPPORT:
WLAN 2: A wireless router provides network connectivity to:**PC1, PC2, PC3, and PC4**: Four computers connected wirelessly.

PRESIDENT ROOM:
LAN 1: A wired local area network established using a switch. It connects:
PC1, PC2, and PC3: Three computers connected via Ethernet cables for high-speed and stable connectivity.
CYBER SECURITY CENTER:
WLAN 1: A wireless router provides connectivity to:
PC1, PC2, PC3, and PC4: Four computers connected wirelessly.

FLOOR 2

FACULTY ROOM:
Faculty room is consisted with 4 pcs and those are connected to WLAN.

CLASSROOM:
In this room 5 pcs are connected to a switch for sharing resources.

GRADUATE ADMISSION:
In this there a 3 pcs connected to WLAN.

3. Network Features

- **Mixed Connectivity**:
  - Wired (LAN) connections in the LAN 1 room ensure high performance for fixed devices.
  - Wireless (WLAN) connections in other rooms provide flexibility for portable devices and ease of deployment
  -

All classrooms and rooms are linked to the central hub in Maxy Hall via network cables, ensuring a cohesive and unified network backbone.

Connection Between Components

1. **Maxy Hall to Rooms**:
   a. Ethernet cables connect the central switch in Maxy Hall to:
      i. **WLAN 2** in the Team Support Room.
      ii. **LAN 1** in the LAN Room.
      iii. **WLAN 1** in the Crisis Security Center.
2. **Intra-Room Connections**:
   a. Wired connections in LAN Room use a switch to link multiple devices.
   b. Wireless routers (WLAN 1 and WLAN 2) enable device connectivity in their respective rooms.

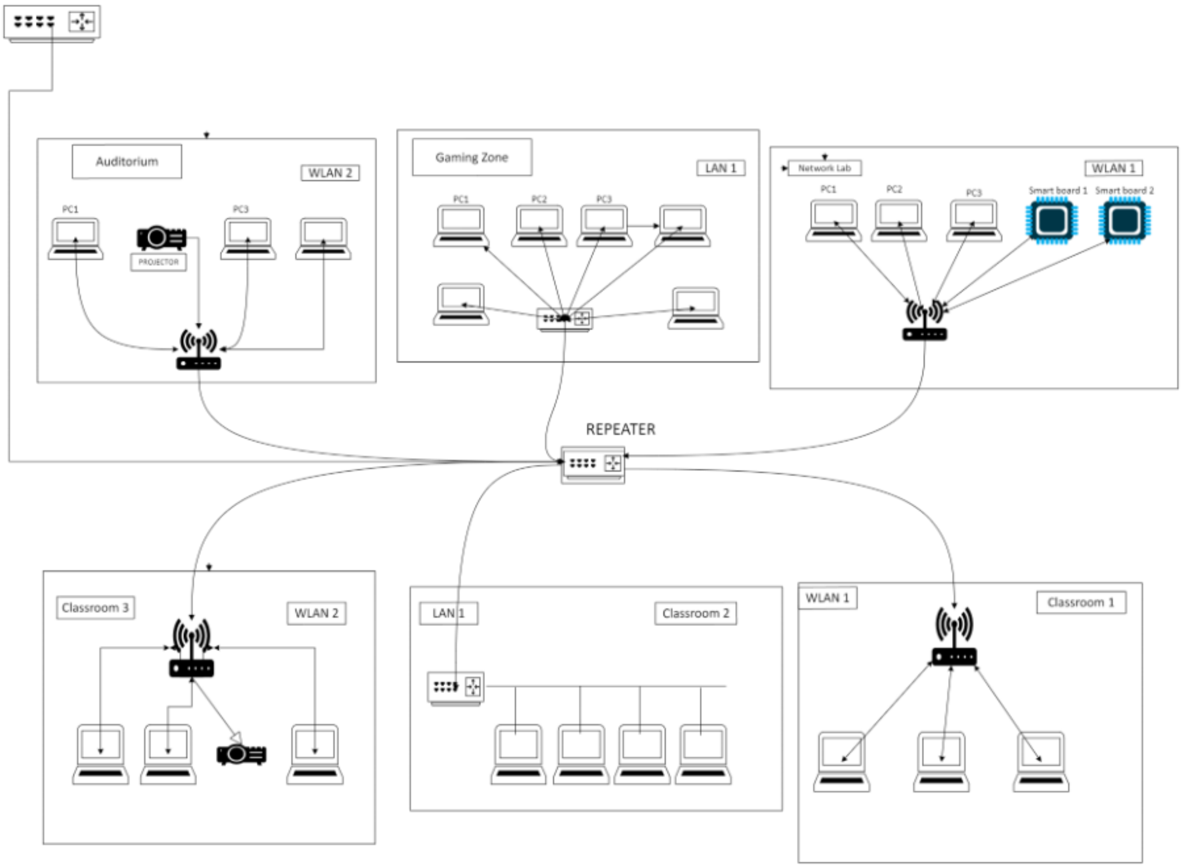
Key Features of the network design

1. **Centralized Management**:
   a. All connections converge at Maxy Hall, making network management and troubleshooting centralized.
2. **Key Features of the network design Mixed Network Types**:
   a. Wired (LAN) for high-performance and reliable connections.
   b. Wireless (WLAN) for flexibility and mobility in device usage.

column1 COMPONENTS	column2 MANUFACTURER	column3 MODEL	column4 COST(APRX)	column5 NO OF UNITS	column 6 TOTAL COST
		TECH SUPPORT			
PC	Apple	imac i7	$1245	5	$6225
PC	HP	Envy	$999	7	$6993
PC	Lenovo	Legion	$879	8	$7032
Wifi	Cisco	CW9166	$1102	1	$1102
		PRESIDENT ROOM			
PC	HP	Envy	$999	6	$5994
PC	Apple	imac i5	$1199	5	$5995
pc	Dell	Inspirion	$799	9	$7191
Switch	Cisco	C1000-8p-2G-L	$367	1	$367
cables	Optospan	PFPF-JM312NXRCD	$2.04/FEET	1000FEET	$2040
		CYBER SECURITY CENTRE			
PC	Apple	imac i7	$1245	8	$9960
PC	HP	Envy	$999	6	$5994
PC	Lenovo	Legion	$879	6	$5274
PC	Dell	Inspirion	$799	9	$7191
Wifi	Cisco	CW9166	$1102	1	$1102
		FACULTY ROOM			
pc	HP	Omen	$1049	8	$8392
pc	HP	Pavilion	$919	9	$8271
pc	DELL	XPS	$899	6	$5394
PC	Lenovo	Legion	$879	6	$5274
wifi	CISCO	CW9166	$1102	1	$1102
		CLASSROOM			
pc	Dell	inspiron	$799	7	$5593
pc	HP	Pavillon	$919	8	$7352
pc	Lenovo	Legion	$879	6	$5274
pc	Lenovo	Idealpad	$829	9	$7461
PC	Apple	imac i5	$1199	3	$3597
switch	Cisco	C1000-8p-2G-L	$367	1	$367
cables	Optospan	PFPF-JM312NXRCD	$2.04?FEET	1000FEET	$2040
		GRADUATE ADMISSION			
pc	Hp	Pavillion	$919	8	$7352
pc	Apple	Imaci7	$1245	7	$8715
pc	Dell	xps	$899	4	$3596
PC	Lenovo	Idealpad	$829	4	$3316
wifi	Cisco	CW9166	$1103	1	$1103
Repetor	Motorla	SLR5100	$2695	1	$2695
TOTAL					$152163

BUCKMAN HALL:

Logical design of Buckman Hall



AN OVERVIEW OF BUCKMAN HALL

This network design ensures seamless connectivity across various areas in a campus or organization. It includes both wired and wireless setups for different zones like classrooms, an auditorium, and a gaming zone.

FLOOR 1.

So as this diagram represents Buckman hall with floors and each floor has represented class rooms.

Auditorium:

In 1st floor there is auditorium with PCs and a projector in it.They are connected to WLAN. WLAN stands for Wireless Local Area Network, which is a network that connects devices using radio frequency to transmit data to share resources.

Gaming Zone:

As gaming zone needs high speed so there is switches for high speed gaming. There are many PCs and laptops for gaming connected to switch. Switch is a hardware device that connects devices on a computer network. It uses packet switching to receive and forward data to the destination device.

Network Lab:

Coming to a room called network lab.There are some PCs and smart boards connected to WLAN.

FLOOR 2.

Classrooms: Computers linked via WLAN and LAN setups for academic use.

Classroom 1: some PCs and a projector in it.They are connected to WLAN. WLAN stands for Wireless Local Area Network, which is a network that connects devices using radio frequency to transmit data to share resources.

Classroom 2: Pcs are connected to a switch to share resources.

Classroom 3: pcs are connected to WLAN.

All zones connect through a central repeater, which relays signals and ensures strong communication across the network. The repeater acts as the backbone, connecting all zones for unified network performance.

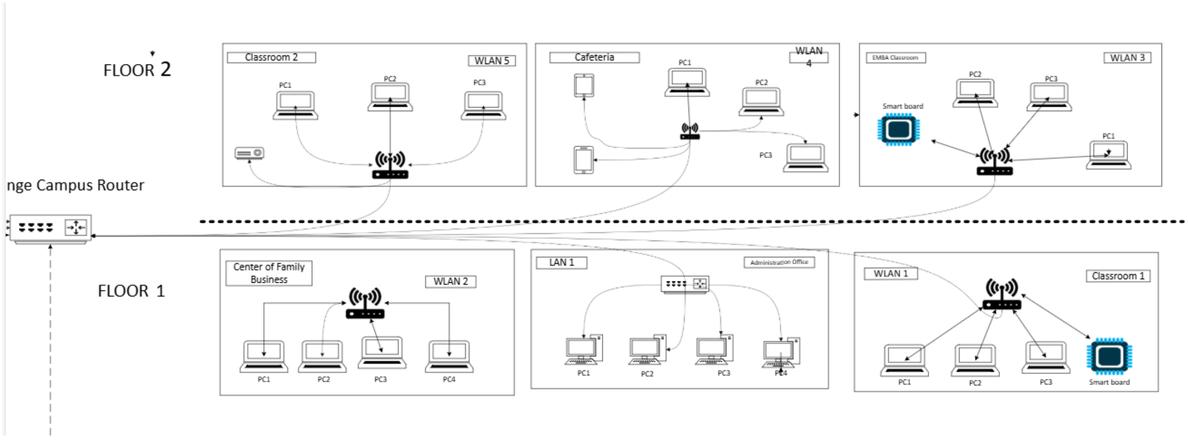
Key Features:

- WLAN (Wireless LAN): Wireless connections in multiple zones for flexibility.
- LAN (Local Area Network): Reliable wired connections for high-speed data transfer.
- Switches and Access Points: Enhance connectivity by managing traffic efficiently.
- Devices are connected in different topologies and in each zone communicate through their local network (wired or wireless).

BUDGET FOR BUCKMAN HALL:

column1 COMPONENTS	column2 MANUFACTURER	column3 MODEL	column4 COST(APRX)	column5 NO OF UNITS	column 6 TOTAL COST
		AUDITORIUM			
PC	Apple	imac i7	$1245	5	$6225
PC	HP	Envy	$999	7	$6993
PC	Lenovo	Legion	$879	8	$7032
Projector	Epson	LS800	$749	1	$749
Wifi	Cisco	CW9166	$1102	1	$1102
		GAMING ZONE			
PC	HP	Envy	$999	6	$5994
PC	Apple	imac i5	$1199	5	$5995
pc	Dell	Inspirion	$799	9	$7191
pc	Apple	Imac i7	$1245	5	$6225
pc	Lenovo	Legion	$879	8	$7032
PC	Lenovo	Legion	$879	7	$6153
Switch	Cisco	C1000-8P-2G-L	$367	1	$367
Cables	Optospan	PFPF-JM312NXRCD	$2.04/Feet 1000 f	1000FEET	$2040
		NETWORK LAB			
PC	Apple	imac i7	$1245	8	$9960
PC	Lenovo	Legion	$879	6	$5274
Smartboard	vibe	S155	$2899	1	$2899
Smartboard	vibe	S155	$2899	1	$2899
PC	Dell	Inspirion	$799	9	$719
Wifi	Cisco	CW9166	$1102	1	$1102
		CLASSROOM1			
pc	HP	Omen	$1049	8	$8392
pc	HP	Pavilion	$919	9	$8271
pc	DELL	XPS	$899	6	$5394
wifi	CISCO	CW9166	$1102	1	$1102
		CLASSROOM2			
pc	Dell	inspiron	$799	7	$5593
pc	HP	Pavillon	$919	8	$7352
pc	Lenovo	Legion	$879	6	$4974
pc	Lenovo	Idealpad	$829	9	$7461
switch	Cisco	C1000-8p-2G-L	$367	1	$367
cables	Optospan	PFPF-JM312NXRCD	$2.04/feet 1000feet	1000FEET	$2040
		CLASSROOM3			
pc	Hp	Pavillion	$919	8	$7352
pc	Apple	Imaci7	$1245	7	$8715
pc	Dell	xps	$899	4	$3596
projector	Epson	LS800	$749	1	$749
wifi	Cisco	CW9166	$1103	1	$1103
Repetor	Motorla	SLR5100	$2695	1	$2695
TOTAL					$1,61,107

ORANGE CAMPUS:



This Image provides a detailed overview of the Orange campus network topology for a two-floor facility. The network includes both wired and wireless components, designed to support communication across classrooms, administrative offices, and common areas. The system is centrally managed through a main Campus Router and comprises various sub-networks for efficient resource allocation and security.

Network Overview:

1.Infrastructure Type: Hybrid (Wired and Wireless)

2.Number of Floors: 2

3.Main Components:

- Campus Router (centralized internet gateway)
- Wireless Access Points (WLANs)
- Ethernet Switches (for wired LANs)
- End-user devices (PCs, tablets, smart boards)

- **Wireless Access Points (WLANs)**: Each WLAN is dedicated to a specific area, ensuring localized wireless connectivity.
- **Switches**: Used in areas requiring high-speed and reliable wired connections, like the Administration Office.
- **Smart Boards**: Integrated into classrooms to support modern teaching methods, connected via wireless networks.
- **Security Considerations**: Implement VLANs or network segmentation to secure communication between floors and departments.

Connectivity:

1. Campus Router

- Acts as the central hub for network traffic, managing internet access for all floors and departments.

- Connects all switches and wireless access points for unified communication.

2.WALN Configuration

- **Access Points**: Strategically placed to ensure strong signal coverage.
- **Separate Networks**: Each area has its dedicated WLAN to avoid congestion and improve performance.

3. Wired LANs

- **Purpose**: Wired connections (e.g., Administration Office) are used for high-speed, low-latency tasks.
- **Switch Configuration**: Ensures reliable and secure data transfer between endpoints.

Benefits

- **Scalability**: Supports the addition of new devices without significant network redesign.
- **Reliability**: Combination of wired and wireless networks ensures redundancy.
- **Security**: VLANs or network segmentation recommended for isolating sensitive traffic.
- **Smart Devices Integration**: Smart boards enhance the learning experience in classrooms.

Column1	Column2	Column3	Column4	Column5	Colum
Components	Manufacture	Model	Cost(appx)	No.of Units	Total Cost
PC	HP	ENVY	$999	10	$9,990
PC	DELL	XPS	$899	5	$4,495
PC	APPLE	MACi9	$1,345	1	$1,345
ROUTER	CISCO	CW9166	$1,102	4	$4,408
SMART BOARD	VIBE	S155	$2,899	2	$5,798
SWITCH	CISCO	C1000-8P-2G-L	$370	1	$370
PROJECTOR	EPSON	LS800	$749	2	$749
TAB	APPLE	10TH GEN	$299	1	$299
CABLE	GENERIC	CAT6 ETHERNET	$5	200	$1,000
TOTAL					$28,454

Total Cost for the proposed network:

Column1	Column2
Building/ Area	Cost
Connection between two campuses	**$801608**
Macy Hall	$152163
Echlin Hall	$23472
Buckman Hall	$161107
Orange Campus	$28454
Total	$1166804

Conclusion:

In this documentation, we can conclude that we have made a comprehensive network design for a "University Campus Area Network" that is inspired from "University Of New Haven". Here we have learned how to go step-by-step from the beginning of the business goals and needs till the final physical diagram. During this process, we also learned how to identify the business goals, technical goals, design a logical network design and to make a physical network design. We also learned how to choose the right components required for making a network and budgeting everything according to the user's demands.